

Chirayata Bhattacharyya

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EDUCATION

Indian Institute Of Science

PG-II, CGPA - 8.80, M. Tech Signal Processing, Electrical Engineering

Bengaluru, India

July 2023 - June 2025

Jadavpur University

CGPA - 9.37, Class Rank - 3/ 68, Bachelor of Engineering in ETCE

Kolkata, India

July 2015 - June 2019

BT Road Govt. Sponsored HS School

Percentage - 95.80, WB State Rank - 12/ 0.65 Mn, Higher Secondary Examination

Kolkata, India

July 2013 - June 2015

Ramakrishna Mission Ashram, Baranagore

Percentage - 92.00, WB State Rank - 36/ 0.9 Mn, Secondary Examination

Kolkata, India

July 2007 - June 2013

INTERESTS & SKILLS

Research: Stochastic Systems, ML Efficiency, Representation Learning, Image & Bio-med Signal Processing

Courseworks: Random Process (A), Linear Algebra (A), Optimisation (A), Pattern Recognition and Neural Networks (A), Deep Representation Learning, Image Processing (A), Detection and Estimation Theory (A)

Languages and Tools: Python, C, Matlab, Mathematica, Git, Linux

Libraries and Packages: PyTorch, Keras, Sklearn, Pandas, Numpy, OpenCV, NLTK, Matplotlib, DataTrove

ML Architectures: Naive Bayes, SVM, CNN, VAE, GAN, Graph NNs, RNN, LSTM, BERT, Transformers, Boosting

RESEARCH EXPERIENCE

■ Research Assistant, Spectrum Lab, EE, IISc Bengaluru, India

Advisor - Dr. Chandra Sekhar Seelamantula, EE, IISc

October 2024 - June 2025

- **Worked** on self-supervised Computational Imaging problems using unbiased estimators, variational calculus and Langevin Dynamics. Worked on several other Inverse Problems across natural, synthetic images and textures.

■ IUSSTF-Viterbi Research Fellow, USC Viterbi School of Engineering

Advisor - Dr. Peter Beerel, ECE, University of Southern California, LA, USA

May 2024 - July 2024

- **Improved 50% Latency** in video object detection using region skipping with Transformers/CNNs as a backbone.
- **Analyzed** effects of Low Rank Adaptation on LLM trust dimensions and mitigation strategies against Jailbreaking.

■ Researcher, ETCE, Jadavpur University

Advisor - Dr. Shankha Sanyal, Sir CV Raman Center for Physics and Music, Kolkata

July 2017 - June 2020

- **Developed** algorithms to model the Mathematical quantification of correlation between emotion, classical Indian Ragas and recitation using Non-linear dynamics (MFDDFA) and Machine Learning, aiming to aid Music Therapy.

WORK EXPERIENCE

■ Senior Engineer, Qualcomm

Visual Tech ML RnD, Qualcomm BDC, Bengaluru

Jan 2026 – Present

- **Dynamic BnR** creating automated parameter predictor for differential Bayer Noise Reduction module in camera.
- **Exploring** scopes of state space models like Vision-Mamba in temporal filtering module, MAC, inside camera ISP.

■ Member of Technical Staff, Cerebras Systems

ML Engineering, Cerebras, Bengaluru

July 2025 – Dec 2025

- **Designed** multi-LORA bring-up; Curated, deduplicated, decontaminated math data at volume across benchmarks.
- **Designed** end-to-end agentic pipeline flow for assembly code generation, validation, scoring and prompt fixation.

■ Research Engineer, TCS Research

Embedded Systems and Robotics, TCS R&I, Kolkata

July 2019 – June 2021

- **Developed** novel algorithms to detect abnormalities in various bio-sensor data; algorithms involved extensive Signal Processing and Pattern Recognition for building prediction systems. **Published** 5 papers, 2 patents.

TEACHING EXPERIENCE

(Responsibilities of TA-ship involved taking tutorials every week, checking copies, arranging doubt-solving sessions for 70-90 students)

- TA, **Random Processes**, Prof. Anurag Kumar and Prof. Rajesh Sundaresan, ECE, IISc [Course Page](#) Fall'24
- TA, **Detection and Estimation Theory**, Prof. Vaibhav Katewa, ECE, IISc [Course Page](#) Spring'25

PUBLICATIONS

- MaskVD: Region Masking for Efficient Video Object Detection *IEEE CVF WACV'25* [Link](#)
- Sepsis Prediction using Continuous and Categorical Features on Sporadic Data, *IEEE Percom'20* [Link](#)
- Language Independent Emotion Quantification using Non linear Modelling of Speech, *JoIPRP'21* [Link](#)
- A Novel Non-Parametric Approach of Tremor Detection Using Wrist-Based PPG Sensor, *IEEE Eusipco'22* [Link](#)
- A Novel Real-time Robust Estimation of Breathing Rate from PPG using Smart Devices. *ACM Sensys'20* [Link](#)

PROJECTS

- **VarSURE: Consistency-driven Regularization for Image Restoration** — Derived a consistency-driven variational SURE-based loss for *unsupervised* denoising; achieved ~ 1 dB PSNR gains over baselines across benchmarks.
- **Scaffold: Insight into Stochastic Controlled Averaging for Federated Learning** - Examined limitations of FedAvg on unbalanced distributed dataset and how SCAFFOLD mitigates client drift reducing data variance.
- **SparseVD: Region Masking for Efficient Video Object Detection** — Reduced inference latency by $\sim 10\%$ using region-skipping with Transformer backbones while preserving detection quality; **accepted at WACV 2025**. [Link](#)
- **The Curious Case of Image-Denoiser Behaviour on Textures** - Benchmarked SOTA restoration models on texture using local metrics (e.g., STSIM variants) and demonstrated a performance gap. **Submitted to SPCOM'26**.
- **Profiling LLM Datasets: Where Does Model Bias Come From?** - Estimated dataset fairness via common-noun geography to assess representativeness and built a generalized auditing framework.
- **OCTCheck: Denoising OCT Images Blindly** - Developed a deep noise-removal method targeting multiplicative noise in OCT using an $N + 2N$ architecture; generates clean images from real-world observations only.
- **Fulcrum Rebalancing and Hybrid Classification for Multi-labeled ECG** - Built a CNN + feature-engineered classifier for multilabel, multiclass problem of 27 ECG classes; addressed class imbalance, ranked 7th globally.
- **LOrthoReD: Learning Orthogonal Transformation for Regularization Using Denoising** - Designed contractive non-linear deep denoisers via PnP-FBS unrolling on wavelet manifolds to improve performance and stability.
- **Breathing Rate Estimator from PPG Data** - Engineered an algorithm on smartwatch PPG signals, leveraging respiration-induced modulations to estimate breathing rate. Prepared a dashboard for real time monitoring.

HONOURS & AWARDS

(x/y denotes x of y recipients selected)

- Received (1/15 India-wide) the **IUSSTF-Viterbi Fellowship'24** to spend 8 weeks at USC as Visiting Researcher [Link](#)
- Awarded with the **Reliance Foundation Post Graduate Scholarship'24** as 1/100 brightest students nationwide.
- Awarded the **IISc-ICASSP Fellowship** to attend IEEE ICASSP'25 at Hyderabad with full registration.
- **UG and PG Entrance Test Scores:** CAT'21 - 99.53 Percentile, GATE'23 - 99.48 Percentile, TIFR GS'19 - AIR 6
- Awarded with the INSPIRE and National Talent Search Examination scholarships by MHRD, Govt. of India.
- Several Times Winner, Inter-college creative writing competitions held at Jadavpur University, ISI Kolkata, '15-'19

EXTRACURRICULARS & PORs

- Convenor, 'Spandan', the Bengali Cultural Organisation of IISc. Anchored and managed annual events end-to-end.
- Actor, 'Rangmanch', the Dramatics Club of IISc. Performed at the Institute at Inter IISc-IISER Cultural Meet'24
- Co-ordinator, 'InnovateX', An AI Hackathon. Managed 50+ teams, Budget INR 2 Lakhs, IISc Rhapsody'25
- Co-ordinator, Poetry Slam 'Mushayera'. Managed 30+ individuals, Budget INR 50k, IISc Rhapsody'24
- Editor, 'Homebound', Cyclone-fund raising webzine'21 and 'Probash', Magazine of IISc Sharodiya Durgotsab'24
- Speakers' Comm. Head, TEDx Jadavpur University'19; Co-ordinator, JU Photography Club, 2017-19